

Hands-on Lab: Getting Started with GitHub

Effort: 20 min

In this lab, you will get started with GitHub by creating a GitHub account and project and adding a file to it using its Web interface.

Objectives

After completing this lab, you will be able to:

0. Describe GitHub
1. Create a GitHub account
2. Add a Project / Repo
3. Edit / Create a file
4. Upload a file & Commit

GitHub Overview

First, let us introduce to GitHub. GitHub in simple words is a collection of folders and files. It is a Git repository hosting service, but it adds many of its own features. While Git is a command-line tool and a server needs to be hosted and maintained via command line as well, GitHub provides this Git server for you and a Web-based graphical interface. It also provides access control and several collaboration features, such as wikis and basic task management tools for every project. GitHub provides cloud storage for source code, supports all popular programming languages, and streamlines the iteration process. GitHub includes a free plan for individual developers and for hosting open source projects.

Exercise 1: Creating a GitHub Account

Please follow the steps given below to create an account in GitHub:

Step 1: Create an account: <https://github.com/join>

NOTE: If you already have a GitHub account, you can skip this step and simply login to your account.

Step 2: Provide the necessary details to create an account as shown below:

Join GitHub

Create your account

Username *

Email address *

Password *

Make sure it's at least 15 characters OR at least 8 characters including a number and a lowercase letter.
[Learn more.](#)

Email preferences

☒ Send me occasional product updates, announcements, and offers.

Verify your account

Please solve this puzzle so we know you are a real person

Verify

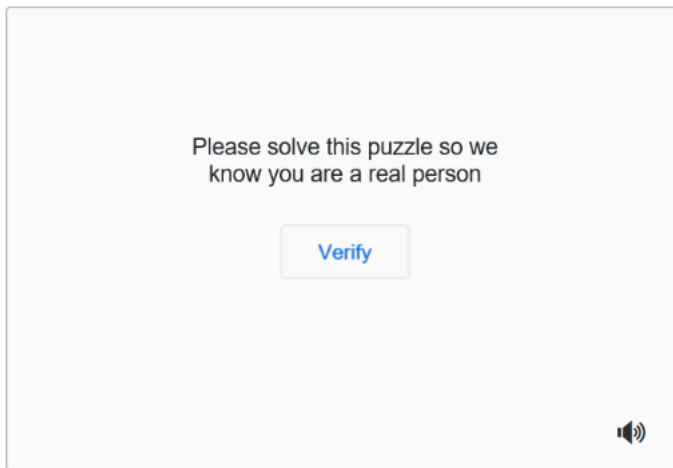
Create account

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

and click **Create account**.

Step 3: Click **Verify** to verify the account and click **Done**

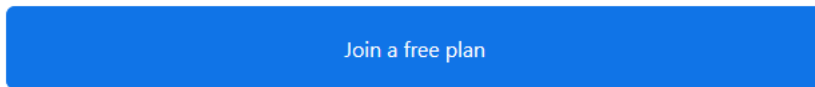
Verify your account



Step 4: After verification, click **Join a Free Plan**

Email preferences

☒ Send me occasional product updates, announcements, and offers.



Step 5: Select the details as shown below and click **Complete Setup**

What do you plan to use GitHub for?

(Select up to 3)

 Learn to code	 Learn Git and GitHub	 Host a project (repository)
 Create a website with GitHub Pages	 Collaborating with my team	 Find and contribute to open source
 School work and student projects	 Use the GitHub API	 Other

I am interested in:

We'll connect you with communities and projects that fit your interests.

For example:

Complete setup

Step 6: Go to your email, find the verification email from GitHub, and click on the link/button in that email to verify your email.

NOTE: If you do not receive verification email, click [Resend verification email](#).



Please verify your email address

Before you can contribute on GitHub, we need you to verify your email address.

An email containing verification instructions was sent to **Your email address**

[Resend verification email](#)[Change your email settings](#)

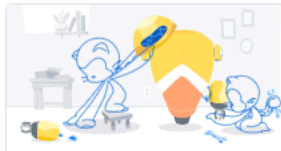
Email is verified

Your email was verified.

✕

What do you want to do first?

Every developer needs to configure their environment, so let's get your GitHub experience optimized for you.



Start a new project

Start a new repository or bring over an existing repository to keep contributing to it.

[Create a repository](#)

Collaborate with your team

Improve the way your team works together and get access to more features with an organization.

[Create an organization](#)

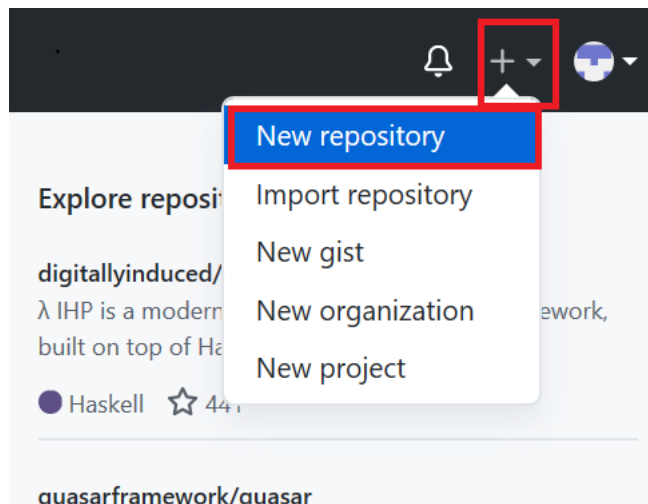
Learn how to use GitHub

Get started with an "Introduction to GitHub" course in our Learning Lab.

[Start Learning](#)[Skip this for now >](#)

Exercise 2: Adding a Project / Repo

Step 1: Click on the + symbol and click New repository.



Step 2: Provide a repository a name and initialize with the empty README.md file.

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Owner *

 Malika-s ▾

Repository name *

/ testrepo ✓

Great repository names are short and memorable. Need inspiration? How about **urban-octo-waffle?**

Description (optional)

☒  **Public**

Anyone on the internet can see this repository. You choose who can commit.

☐  **Private**

You choose who can see and commit to this repository.

Skip this step if you're importing an existing repository.

☒ **Initialize this repository with a README**

This will let you immediately clone the repository to your computer.

Add .gitignore: None ▾

Add a license: None ▾



Create repository

and click `Create repository`.

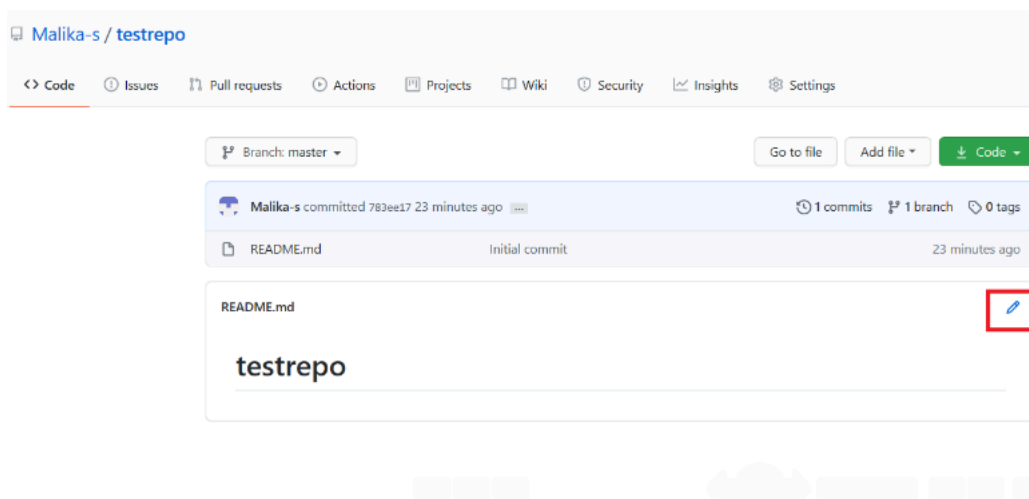
Now, you will be redirected to the repository you have created.

Let's start editing the repository.

Exercise 3: Create / edit a file

Exercise 3a: Edit a file

Step 1: Once the repository is created, the root folder of your repository is listed by default and it has just one file `ReadMe.md`. Click on the pencil icon to edit the file.



Malika-s / testrepo

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Branch: master ▾ Go to file Add file ▾ Code ▾

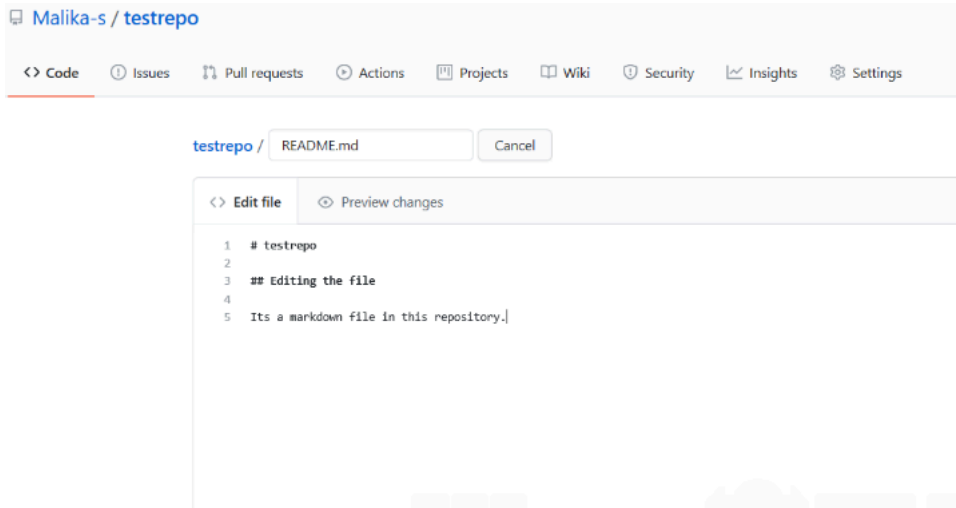
Malika-s committed 783ee17 23 minutes ago 1 commits 1 branch 0 tags

README.md Initial commit 23 minutes ago

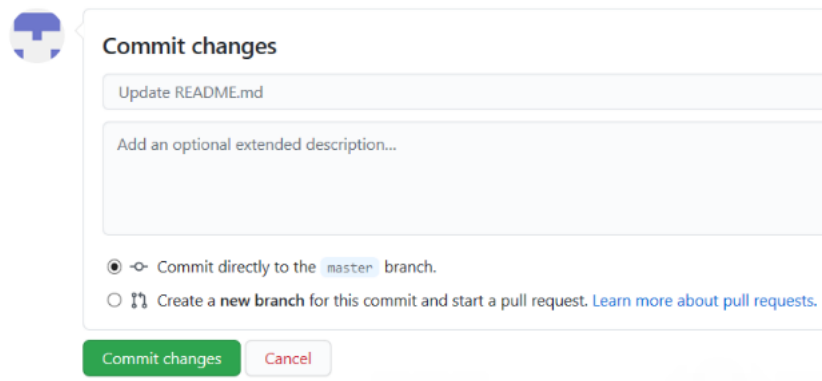
README.md

testrepo

Step 2: Add text to file.



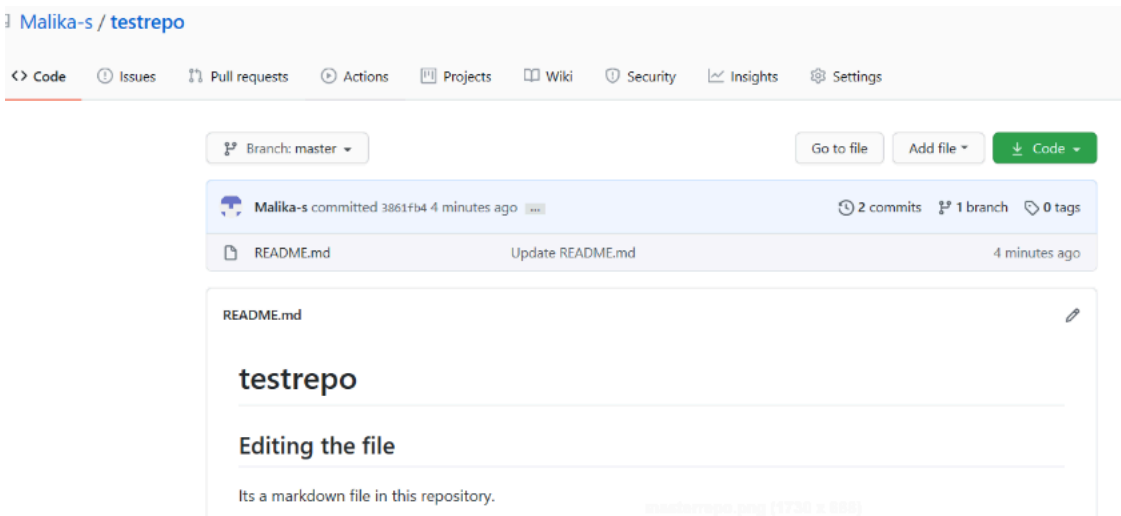
Step 3: Scroll down the page after adding the text and click commit Changes.



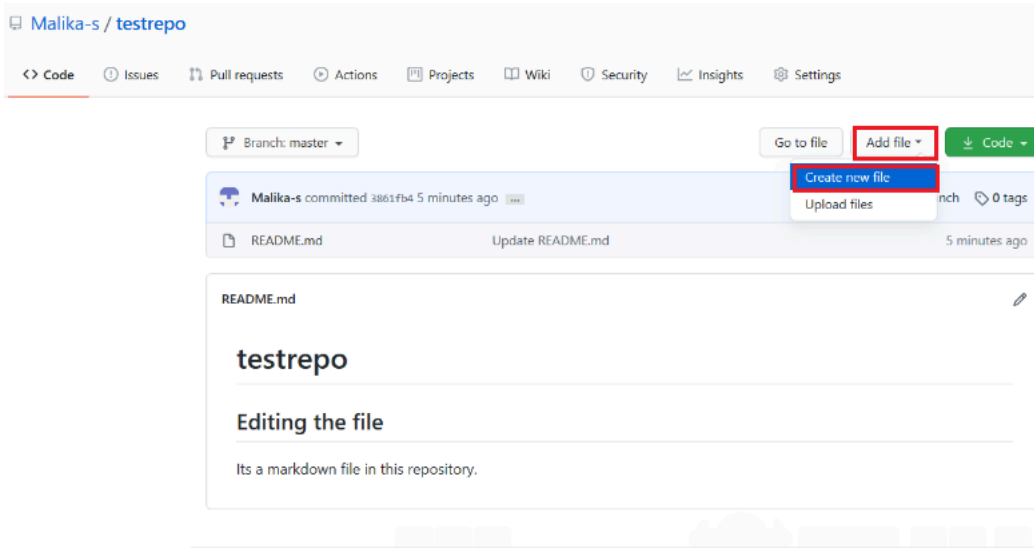
Now, check your file is edited with the new text.

Exercise 3b: Create a new file

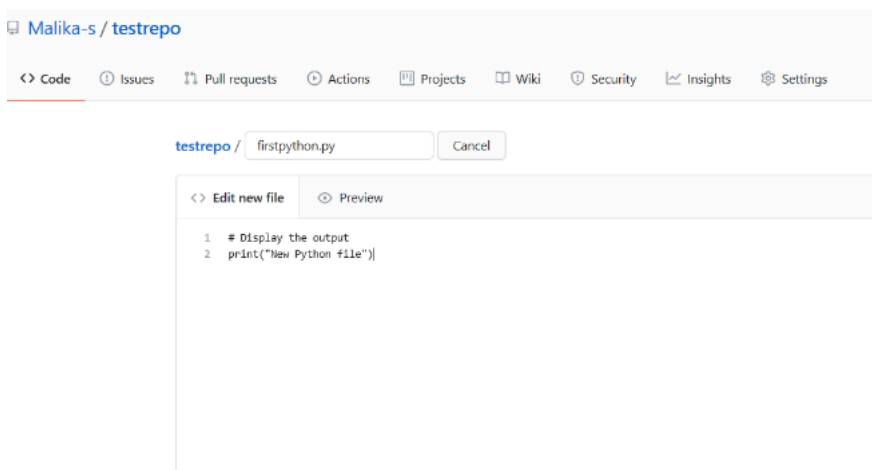
Step 1: Click on the repository name to go back to the master branch like in this testrepo.



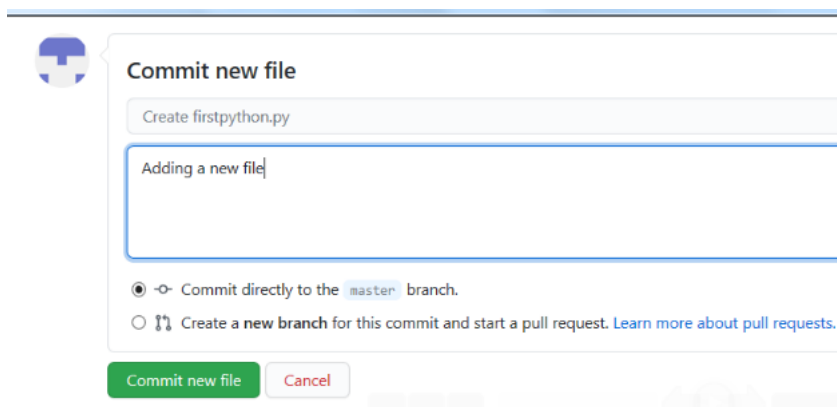
Step 2: Click Add file and select Create New file to create a file in the repository.



Step 3: Provide the file name and the extension of the file. For example, firstpython.py and add the lines.



Step 4: Scroll down the page after adding the text. Add description of the file (optional) and click commit new file.



Step 5: Your file is now added to your repository and the repository listing shows when the file was added/changed.

Exercise 4: Upload a file & Commit

Step 1: Click Add file and select upload files to upload a file (Upload any .txt,.ipynb, .png file) in the repository from the local computer.

The screenshot shows the GitHub interface for a repository named 'testrepo' by user 'Malika-s'. The top navigation bar includes links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. Below the navigation bar, there's a 'Branch: master' dropdown and buttons for 'Go to file', 'Add file', and 'Code'. The 'Add file' dropdown menu is open, showing options for 'Create new file' and 'Upload files'. The 'Upload files' option is highlighted with a red box. Below the menu, a list of files is shown: 'README.md' (Update README.md, 1 hour ago) and 'firstpython.py' (Create firstpython.py, 1 hour ago). At the bottom, the 'README.md' file content is visible.

Step 2: Click on choose your files and choose any files from your computer.

The screenshot shows the GitHub interface for the 'testrepo' repository. It features a large box with the text 'Drag files here to add them to your repository' and a link 'Or choose your files'. Above the text are icons representing different file types: a document, a folder, a code file, a list, and a PDF.

Step 3: Once the file finishes uploading, click on commit Changes

The screenshot shows the GitHub 'Commit changes' dialog box. At the top, the file 'Test.py' is listed and highlighted with a red box. Below the file list, there's a section titled 'Commit changes' with a text area for 'Add an optional extended description...'. Underneath, there are two radio buttons: 'Commit directly to the master branch.' (selected) and 'Create a new branch for this commit and start a pull request.' At the bottom, there are two buttons: 'Commit changes' (highlighted with a red box) and 'Cancel'.

Step 4: Now, your file is uploaded in the repository.

The screenshot shows the GitHub repository 'testrepo' after the file upload. The 'Add file' dropdown menu is open, showing the 'Upload files' option. Below the menu, a list of files is shown: 'README.md' (Update README.md, 2 hours ago), 'Test.py' (Add files via upload, now), and 'firstpython.py' (Create firstpython.py, 1 hour ago). The 'Test.py' file is highlighted with a red box.

Summary

In this document, you have learned how to create a new repository, adding a new file, editing a file, and uploading a file in a repository and commit the changes.

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